Physical diffusion models with fast latent units: model, local training and measured statistics

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Abstract

We extend the physical diffusion model — a lattice of nonlinear (Duffing) oscillators, one per pixel, whose energy is trained per time slice by score matching — with latent degrees of freedom: hidden units that couple to the pixels and equilibrate much faster than they do. Integrated out, the hidden units give the pixels a free energy with a global, many-body part, in particular a "which prototype" variable that a two-body pixel energy lacks. We derive the free energy for Ising, Potts and per-pixel hidden units, the score-matching objective and its gradient, and show that the gradient is a local three-factor rule. On a device the hidden statistics must be measured from a few samples; we derive the bias this causes, remove it with a first- and second-order jackknife, and describe an optimiser (the visit-count rule) that keeps rarely visited hidden states alive. We close with the reverse (generative) process, the physical conditions for the fast-hidden idealisation, the evaluation, and the main numerical results so far.

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1 Setting: the physical diffusion model

Forward process. The data $\boldsymbol{x} \in \mathbb{R}^N$ (pixels, $N = 784$ at 28×28, $N = 196$ at 14×14, normalised) are diffused by an Ornstein–Uhlenbeck process at temperature T,

$$d\boldsymbol{x} = -\boldsymbol{x}\, dt + \sqrt{2T}\, d\boldsymbol{W}_t, \qquad T = 0.02, \qquad t \in [0, 5], \tag{1}$$

whose stationary density is $\propto e^{-\|\boldsymbol{x}\|^2/2T}$. The marginal at time t is p_t.

Time slices. The interval $t \in [0, 5]$ is cut into 100 slices. Every parameter of the model has its own value in each slice; slice k is trained on fresh samples of p_{t_k} and warm-started from slice $k-1$. During generation the parameters are interpolated between the slice knots.

Energy-based score. At each slice the model density is $p_\theta(\boldsymbol{x}) \propto e^{-F_\theta(\boldsymbol{x})/T}$, so its score is $\nabla \log p_\theta = -\nabla F_\theta / T$. Without hidden units $F_\theta = E^{\mathrm{v}}_\theta$, the visible energy.

Visible energy. One oscillator per pixel, with self terms and pair couplings,

$$E^{\mathrm{v}}(\boldsymbol{x}) = \sum_i \Big[\tfrac{k_i^{\mathrm{lin}}}{2} x_i^2 + \tfrac{k_i^{\mathrm{duff}}}{4} x_i^4 + \tfrac{k_i^{(6)}}{6} x_i^6 + \beta_i x_i \Big] + \sum_{(i,j)\in\mathcal{P}} \Big[c_{ij}^{\mathrm{lin}} (x_i - x_j)^2 + c_{ij}^{\mathrm{om}}\, x_i^2 x_j + \tfrac{c_{ij}^{\mathrm{duff}}}{4} (x_i - x_j)^4 \Big]. \tag{2}$$

It is linear in its parameters θ_v. The paper's model ("power 6") keeps $k_i^{(6)} \geq 0.01$ for confinement; "power 4" sets $k^{(6)} = 0$ and confines with $k_i^{\mathrm{duff}} \geq 0.01$, $c^{\mathrm{duff}} \geq 0$. A pixel's self energy is a double well wherever $k_i^{\mathrm{lin}} < 0$. The pair set $\mathcal{P}$ contains all pairs within a Chebyshev distance (14 at 28×28: 180,810 pairs, 59% of all pairs; 7 at 14×14). The three pair terms are the linear spring (c^{lin}), an asymmetric "optomechanical" term (c^{om}) and a quartic pair term (c^{duff}). Options `--radius`, `--neighbourhood von_neumann` and `--pair-terms` restrict $\mathcal{P}$ and the pair terms (Section 8).

2 Latent degrees of freedom

2.1 Joint energy and marginal

Add hidden units $\boldsymbol{y} \in \mathbb{R}^K$ that couple bilinearly to the pixels and not to each other (a bipartite, RBM-like structure):

$$E_\theta(\boldsymbol{x}, \boldsymbol{y}) = E^{\mathrm{v}}(\boldsymbol{x}) + \sum_{j=1}^{K} \big[U_j(y_j) - y_j\, a_j(\boldsymbol{x}) \big], \qquad a_j(\boldsymbol{x}) = b_j + \boldsymbol{W}_j \cdot \boldsymbol{x}, \tag{3}$$

with U_j the unit's own potential, e.g. a double well. The joint energy is linear in all parameters. Integrating out the hidden units gives the marginal $p_\theta(\boldsymbol{x}) \propto \int d\boldsymbol{y}\, e^{-E_\theta/T} = e^{-F_\theta(\boldsymbol{x})/T}$ with the free energy

$$F(\boldsymbol{x}) = E^{\mathrm{v}}(\boldsymbol{x}) - T \sum_j \log Z_j\big(a_j(\boldsymbol{x})\big), \qquad Z_j(a) = \int dy\, e^{-(U_j(y) - ya)/T}. \tag{4}$$

$T \log Z_j$ is the cumulant generating function of y_j; writing m_j, v_j, $k_{3,j}$ for the conditional mean, variance and third cumulant of y_j given $\boldsymbol{x}$,

$$T \partial_a \log Z_j = m_j, \qquad \partial_a m_j = v_j / T, \qquad \partial_a v_j = k_{3,j} / T. \tag{5}$$

Idealisation. We always integrate the hidden units out exactly. Physically this is the limit of hidden units that equilibrate infinitely fast compared with the pixels (Section 6).

2.2 Ising units

For a deep double well with minima at $\pm y_0$, $y_j \to y_0 h_j$ with $h_j = \pm 1$ (y_0 absorbed into $\boldsymbol{W}, b$):

$$Z_j = 2\cosh(a_j/T), \qquad m_j = \tanh(a_j/T), \qquad v_j = 1 - m_j^2, \qquad k_{3,j} = -2m_j v_j. \tag{6}$$

With K Ising units the model is a mixture of 2^K exponentially tilted copies of the visible model,

$$p_\theta(\boldsymbol{x}) \propto \sum_{\boldsymbol{h}\in\{\pm1\}^K} \exp\Big(- \big[E^{\mathrm{v}}(\boldsymbol{x}) - \textstyle\sum_j h_j a_j(\boldsymbol{x})\big]/T\Big). \tag{7}$$

One unit with $\boldsymbol{W}_1$ along the Fisher (LDA) direction separates two classes; $P(h = +1|\boldsymbol{x}) = \sigma(2a_1/T)$ is then the LDA posterior. So far only $K = 1$ has been trained; several units need a symmetry-breaking initialisation, because $\boldsymbol{W} = 0$ is a degenerate stationary point (Section 3).

2.3 Potts units

A Potts unit is *one* hidden unit with M mutually exclusive states $m = 1 \dots M$ (exactly one active). State m has its own couplings to all pixels and its own field:

$$F(\boldsymbol{x}) = E^{\mathrm{v}}(\boldsymbol{x}) - T\log\sum_{m=1}^{M} e^{a_m(\boldsymbol{x})/T}, \qquad p_m(\boldsymbol{x}) = \frac{e^{a_m/T}}{\sum_k e^{a_k/T}}. \tag{8}$$

The model is a mixture of M tilted copies of the visible model; each state acts as a prototype. "Potts 128" is one unit with 128 states, not 128 units. In hardware a Potts unit is an M-well oscillator, or M bistable units with strong lateral inhibition (winner-take-all); the code idealises it as exactly one-hot.

2.4 Spring coupling and per-pixel units

Spring coupling (`--quad`). The input field also contains the squared pixels,

$$a_m(\boldsymbol{x}) = b_m + \boldsymbol{W}_m \cdot \boldsymbol{x} + \boldsymbol{Q}_m \cdot \boldsymbol{x}^2, \qquad \boldsymbol{G}_m \equiv \nabla a_m = \boldsymbol{W}_m + 2\boldsymbol{Q}_m \circ \boldsymbol{x}. \tag{9}$$

State m then adds $-Q_{mi}x_i^2$ to pixel i's energy: it changes the pixels' stiffness, so the mixture components can differ in width and shape, not only in position. A bilinear coupling alone only shifts the components (for a quadratic E^{v} exactly), so class-specific shape would otherwise have to come from the anharmonic part of E^{v}.

Per-pixel units (`--local`). One Ising unit per pixel, coupled only to its own pixel, $a_i = c_i + w_i x_i$, adds $-T\log 2\cosh\big((c_i + w_i x_i)/T\big)$: an asymmetric double well per pixel (ink or background). "Potts M + both" denotes one Potts unit with spring coupling plus the per-pixel units:

$$F = E^{\mathrm{v}} - T\log\sum_m e^{a_m/T} - T\sum_i \log 2\cosh\frac{c_i + w_i x_i}{T}. \tag{10}$$

2.5 What the latent units add

Mean force and softening. With $\boldsymbol{\psi} \equiv \nabla F$ (the force a clamp must supply to hold the pixels), for Ising units

$$\boldsymbol{\psi} = \nabla E^{\mathrm{v}} - \sum_j m_j \boldsymbol{W}_j, \qquad \nabla\nabla F = \nabla\nabla E^{\mathrm{v}} - \frac{1}{T}\sum_j v_j \boldsymbol{W}_j \boldsymbol{W}_j^{\top}, \qquad \Delta F = \Delta E^{\mathrm{v}} - \frac{1}{T}\sum_j v_j \|\boldsymbol{W}_j\|^2. \tag{11}$$

The hidden units exert the mean force $\sum_j m_j \boldsymbol{W}_j$ and soften the landscape along $\boldsymbol{W}_j$; near $a_j = 0$ a unit turns a single well into a double well along $\boldsymbol{W}_j$ where $\|\boldsymbol{W}_j\|^2/T$ exceeds the visible curvature.

A many-body, global variable. The mode-selection force $W_{1i}\tanh(a_1(\boldsymbol{x})/T)$ on pixel i is a nonlinear function of a global projection of all pixels. A two-body energy cannot represent it exactly: its forces satisfy $\partial_k \partial_l f_i = 0$ for distinct $k, l \neq i$, while the tanh force does not. Expanding, $-T\log\cosh(a/T) = -a^2/2T + a^4/12T^3 - \dots$ contains products of four distinct pixels. The a^2 part is itself a pair term; the hidden units add the non-Gaussian many-body remainder. Empirically, adding more pair couplings to the paper's model (all pixel pairs) does not reduce 0/1 blends, while one hidden unit does.

A linear hidden mode is not enough. A harmonic hidden mode $U(y) = \frac{k}{2}y^2$ integrates out to $-a(\boldsymbol{x})^2/2k$: a rank-one softening, i.e. another pair term, never the $\pm\boldsymbol{W}$ mixture. The hidden unit must be nonlinear (bistable or multi-well).

3 Training: score matching per slice

3.1 Objective

For $p \propto e^{-F/T}$ the implicit score-matching (Hyvärinen) objective of slice t is

$$J_T(\theta) = \mathbb{E}_{\boldsymbol{x}\sim p_t}\Big[-\frac{\Delta F}{T} + \frac{\|\nabla F\|^2}{2T^2}\Big], \qquad \partial_\theta J_T = \mathbb{E}\Big[-\frac{\partial_\theta \Delta F}{T} + \frac{(\partial_\theta \nabla F)\cdot\boldsymbol{\psi}}{T^2}\Big]. \tag{12}$$

It is minimised per slice with Adam on minibatches of 512 fresh samples of p_t; the slice result is an exponential moving average of the iterates. Because F is nonlinear in the hidden parameters, J_T is no longer a convex quadratic; it has gauge symmetries (sign flips, permutations of identical units), and $\boldsymbol{W} = 0$ is a degenerate stationary point (to second order in $\boldsymbol{W}$ the hidden free energy only adds terms that the visible bias and pair couplings absorb). Potts states are therefore initialised from k-means prototypes (or found from a random start, which also works), and the single Ising unit along the Fisher direction.

3.2 Visible parameters

For $E^{\mathrm{v}} = \sum_k \theta_k \phi_k$ the form is unchanged, with $\boldsymbol{\psi}$ now including the hidden mean force:

$$\partial_{\theta_k} J_T = \mathbb{E}\Big[-\frac{\Delta\phi_k}{T} + \frac{\nabla\phi_k\cdot\boldsymbol{\psi}}{T^2}\Big]. \tag{13}$$

3.3 Ising couplings: a local three-factor rule

$$\partial_{W_{ji}} J_T = \frac{1}{T^2}\mathbb{E}\Big[-m_j\psi_i + 2v_j W_{ji} + \frac{x_i}{T}\big(k_{3,j}\|\boldsymbol{W}_j\|^2 - v_j\,\boldsymbol{W}_j\cdot\boldsymbol{\psi}\big)\Big], \tag{14}$$

$$\partial_{b_j} J_T = \frac{1}{T^3}\mathbb{E}\Big[k_{3,j}\|\boldsymbol{W}_j\|^2 - v_j\,\boldsymbol{W}_j\cdot\boldsymbol{\psi}\Big]. \tag{15}$$

Each update needs, at pixel i, its position x_i and clamp force ψ_i; at hidden unit j, the statistics of its own fluctuations (m_j; $v_j, k_{3,j}$ follow for Ising units), $\|\boldsymbol{W}_j\|^2$ and the projection $\boldsymbol{W}_j \cdot \boldsymbol{\psi}$; and the coupling itself. The projection is the change of a_j when the clamped pixels are nudged along $\boldsymbol{\psi}$; because a_j is linear (or quadratic, with $\boldsymbol{Q}$) in $\boldsymbol{x}$, a symmetric pair of nudges $\pm\varepsilon\boldsymbol{\psi}$ gives it exactly. No free-running negative phase is needed.

3.4 Potts couplings with spring coupling

For the model (10), per clamped sample define for state m: p_m, $\boldsymbol{G}_m$, $s_m = \sum_i Q_{mi}$; for the unit: $\bar{\boldsymbol{G}} = \sum_k p_k \boldsymbol{G}_k$, $V = \sum_k p_k \|\boldsymbol{G}_k - \bar{\boldsymbol{G}}\|^2$, $\bar{s} = \sum_k p_k s_k$; and the scalar

$$c_m = \|\boldsymbol{G}_m - \bar{\boldsymbol{G}}\|^2 - V + 2T(s_m - \bar{s}) - (\boldsymbol{G}_m - \bar{\boldsymbol{G}}) \cdot \boldsymbol{\psi}. \tag{16}$$

Then

$$\partial_{b_m} J_T = \frac{\langle p_m c_m \rangle}{T^3}, \qquad \partial_{W_{mi}} J_T = \Big\langle p_m \Big[\frac{x_i c_m}{T^3} + \frac{2(G_{mi} - \bar{G}_i) - \psi_i}{T^2} \Big] \Big\rangle, \tag{17}$$

$$\partial_{Q_{mi}} J_T = \Big\langle p_m \Big[\frac{x_i^2 c_m}{T^3} + \frac{2x_i\big(2(G_{mi} - \bar{G}_i) - \psi_i\big)}{T^2} + \frac{2}{T} \Big] \Big\rangle. \tag{18}$$

Coupler (m, i) needs a state-level scalar (p_m, c_m) times a pixel signal (x_i, ψ_i, $\bar{G}_i$): a three-factor rule. The per-pixel units follow the Ising rule with a single coupling. This local rule equals the automatic gradient of the training loss to relative differences of 10^{-16}–$4 \cdot 10^{-11}$ on trained models: every exactly trained model is trained by it, given exact hidden statistics.

3.5 Contrastive-divergence form

Hyvärinen's relation between one-step contrastive divergence (CD-1) with a short Langevin release and score matching gives $\langle \partial_\theta F(\boldsymbol{x}) \rangle - \langle \partial_\theta F(\boldsymbol{x}') \rangle = \frac{T^2 dt}{2} \partial_\theta J_T$, i.e. a Hebbian term minus the same term after a short release. It trains as well as the exact gradient with exact hidden averages, but with sampled hidden states the difference of two independent readouts drowns the $O(dt)$ signal; force matching is therefore the rule for a device.

4 Training on a device: measured hidden statistics

4.1 Plug-in bias

A device cannot compute p_m or m_j; it reads the hidden units S times while the pixels are clamped and averages. Replacing the exact statistics by these sample means $\hat{p}_m$ ("measured statistics", S samples or *reads* per configuration) gives an unbiased estimate of everything linear in them, but the loss is quadratic and its parameter gradient at most cubic in the sample means of one configuration. Hence, over n draws,

$$\mathbb{E}\big[\hat{L}_n\big] = a + \frac{b}{n} + \frac{c}{n^2}, \qquad \text{in particular} \quad \mathbb{E}\big[\hat{J}_S\big] = J_T - \frac{V}{2ST^2} \tag{19}$$

for the loss, with V the variance of the hidden force: the plug-in estimate is biased at order $1/S$.

4.2 Jackknife corrections

Split the S draws into halves and quarters and combine the estimates from the same draws:

$$\text{first order (jk):} \quad 2\hat{L}_S - \tfrac{1}{2}\big(\hat{L}^{(1)}_{S/2} + \hat{L}^{(2)}_{S/2}\big) \;\Rightarrow\; \mathbb{E} = a - \frac{2c}{S^2}, \tag{20}$$

$$\text{second order (jk2):} \quad \tfrac{8}{3}\hat{L}_S - 2\,\overline{\hat{L}_{S/2}} + \tfrac{1}{3}\,\overline{\hat{L}_{S/4}} \;\Rightarrow\; \mathbb{E} = a. \tag{21}$$

(The coefficients of b/n cancel as $\frac{8}{3} - 2 \cdot 2 + \frac{1}{3} \cdot 4 = 0$, those of c/n^2 as $\frac{8}{3} - 2 \cdot 4 + \frac{1}{3} \cdot 16 = 0$.) To first order in the fluctuations neither adds noise. On a device this means averaging the hidden readouts over one full, two half and four quarter windows. The gradient of the visible parameters is linear in the sample means and the same for every order.

4.3 Optimisers and the loss of hidden states

With few samples, a rarely occupied Potts state is rarely visited: its gradient is zero in most steps and a spike in the others (unbiased with the jackknife). The optimiser decides whether such a state survives:

- **Per-element Adam** divides each coupling's gradient by its own running root-mean-square, so a rarely visited state moves at most about $\sqrt{\lambda}$ as fast as with exact statistics (λ = visits per step). States below about one visit per step fall behind and are lost.
- **Plain steps** (one step size per parameter group) lose most states even with exact statistics, because the gradient of a state carries its occupation.
- **Gain per state** (one adaptive step size per hidden state) keeps them with exact statistics but not with measured ones.
- **Visit-count rule** (`--hidden-norm occupancy`): each state keeps a leaky average of its gradient and a leaky count of its visits; its step is the ratio, under one occupation-weighted step scale per parameter group, clipped so that no parameter moves more than one learning rate per step. Schematically, per state m,

$$\bar{g}_m \leftarrow \beta \bar{g}_m + (1-\beta) g_m, \qquad \bar{n}_m \leftarrow \beta \bar{n}_m + (1-\beta) \hat{p}_m, \qquad \Delta\theta_m \propto -\eta \,\mathrm{clip}\Big(\frac{\bar{g}_m}{\bar{n}_m \, s}, \pm 1\Big), \tag{22}$$

  with $\beta = 0.999$ and s the group scale. Dividing by the visit count removes the occupation factor that makes rare states slow. It needs one counter per state and one leaky average per coupling. The visit counts can be taken from the same S draws as the statistics (`--occ-same-draws`), so it reads no more than per-element Adam.

5 Generation: the reverse process

In reversed time $\tau = 5 - t$ the sampler integrates

$$d\boldsymbol{x} = \big(\boldsymbol{x} - 2\nabla F_{\theta(\tau)}(\boldsymbol{x})\big)\, d\tau + \sqrt{2T}\, d\boldsymbol{W}_\tau, \tag{23}$$

i.e. the pixels move in the effective potential $2F - \frac{1}{2}\|\boldsymbol{x}\|^2$ at temperature T. Without hidden units this is the visible model with parameters $2\theta - \theta_{\mathrm{lin}}$. With hidden units, doubling $\boldsymbol{W}$ and b is *not* enough ($-T \log 2\cosh(2a/T) \neq -2T \log 2\cosh(a/T)$). An exact construction at one temperature is to duplicate every hidden unit (two replicas at T with the trained couplings), which gives $-2T \sum_j \log Z_j$ exactly; alternatives are the doubled energy at hidden temperature $2T$, or an amplified back-action force. In simulation the drift $\boldsymbol{x} - 2\nabla F$ is used directly.

6 When the idealisation holds

The hidden units must follow their input field. Near $t = 0$ the pixel relaxation time in generation is about 0.03 (stiffness $2k - 1 \approx 36$), and pixel noise moves a_j by about T in a time $T/(2\|\boldsymbol{W}_j\|^2)$; the hidden correlation time must be short against both. At equal temperatures (clamped training, releases, replica generation) detailed balance keeps the pixel marginal exactly $e^{-F/T}$ for any hidden correlation time, which then only adds friction; with two temperatures it also heats

the pixels. Bistability and speed pull against each other: a deep barrier gives Ising-like units, a shallow one fast switching (Kramers rate $\propto e^{-\Delta U/T}$); the way out is a much higher attempt frequency than the pixels, with barriers of a few T. A Potts unit built from M bistable units with finite inhibition will sometimes have zero or two units on; whether the model tolerates that is untested.

7 Evaluation

Each generated sample is scored by three flags, calibrated on real images (training images scored leave-one-out): *diverged* (an amplitude above 15 normalised units), *blend* (a 0 and a 1 overlaid: one template of each class fits clearly better than any single template; 5% of real images are flagged) and *far* (its nearest training image is farther than for 90% of real images at 14×14; a fixed distance at 28×28). A sample is *clean* if it is none of these. The *clean yield (50/50)* averages the clean share over the two classes among non-diverged samples; real images score 85.7% (14×14) and 83% (28×28). Variety is measured separately by k-nearest-neighbour precision and recall ($k = 3$) per class. All models are sampled from the same 500 initial states and noise paths, so comparisons are paired (McNemar tests; paired bootstrap, 95% intervals).

8 Results so far

problem	model	clean yield
28×28, 0/1	paper's model (no hidden units)	31.9%
	one Ising unit with spring	43.5%
	Potts 128 + both, exact training	86.3%
	Potts 128 + both, local rule, $S = 64$, jackknife	87.6%
	Potts 128 + both, visit-count rule, $S = 16$	87.1%
	Potts 512 + both, power 4, exact training	92.3%
14×14, 0/1 (8 reads)	Potts 128 + both, exact training	84.3%
	local rule, per-element Adam, $S = 8$	68.4%
	local rule, visit-count rule, $S = 8$, same-draw counts	81.7%
14×14, 0/1, power 4 Potts 128 + both exact training	radius 7 / 3 / 2	84.7 / 83.9 / 83.8%
	8 nearest neighbours, springs only	81.4%
	8 nearest neighbours, + optomechanical term	81.1%
	4 nearest neighbours, springs only	74.2%
	no pair couplings	60.0%
	4 nearest neighbours, springs only, no hidden units	0%
14×14, 0/1/2/3	Potts 512 + both, exact / local ($S = 64$, visit-count)	85.0 / 74.7%

Table 1: Main results (one training run per arm; seed-to-seed variation about one point where measured). Real images score 83% (28×28) and 85.7% (14×14).

- Hidden units fix the paper model's main failure (0/1 superpositions) and, with enough Potts states and the spring coupling, bring the samples to the level of real images; power 4 matches power 6 from about 128 states.
- The score-matching gradient is a local three-factor rule. With measured hidden statistics it needs the jackknife and an optimiser that keeps rare states alive: the visit-count rule beats per-element Adam by 13 points at equal reads (8 per configuration), replicated at a second training seed, and with 16 samples matches exact training at 28×28.

- A much simpler visible model works: nearest-neighbour linear springs (8 neighbours) cost about 3 points; the optomechanical term is not needed. Without hidden units such a model produces no clean digits.
- Open: local training on four digits (-10 points); quality as a function of the number of Potts states and of Ising units with the simplest visible model (running); several Ising units instead of one Potts unit; finite hidden speed and finite winner-take-all inhibition.